

High-Dimensional Space

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Content based on *Foundations of Data Science, Chapter 2*

Avrim Blum, John Hopcroft, and Ravindran Kannan, Thursday 27th February, 2020

Introduction

- ▶ Word Vector Model
- ▶ Webpage Model
- ▶ Nearest neighbor query
- ▶ Information retrieval
- ▶ Online recommendation
- ▶ ...

d -dimensional space: $\mathbb{R}^d$, d is large; unit ball: $B_1 := \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}$

1. The **geometric** property of high-dimensional space
 - the volume of the unit ball vanish as d increase
2. The **probabilistic** behavior of high-dimensional random variable
 - drawing n points from d -dim Gaussian, the distance

Generate random points in $\mathbb{R}^d$ using $N(0, \sigma^2)$ to generate coordinates.

For any two points, the **square of distances** are all large but closed to each other.

Because

$$\mathbb{E}[\|\mathbf{y} - \mathbf{z}\|^2] = \mathbb{E}\left[\sum_{i=1}^d (y_i - z_i)^2\right] = d(2\sigma^2) = O(d), \quad (1)$$

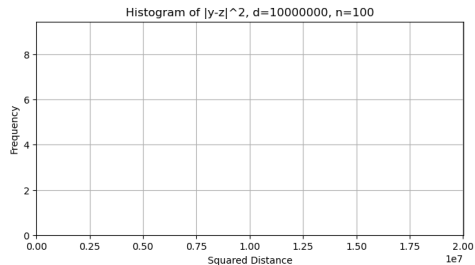
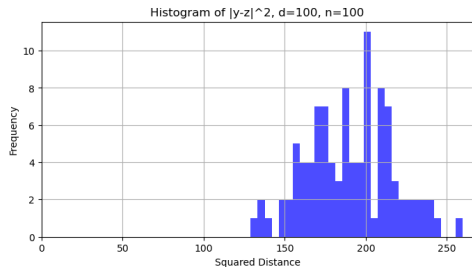
$$sd(\|\mathbf{y} - \mathbf{z}\|^2) = \sqrt{\text{Var}(\|\mathbf{y} - \mathbf{z}\|^2)} = \sqrt{d(8\sigma^4)} = O(\sqrt{d}). \quad (2)$$

We find that the "relative deviation" is $\propto \frac{\sqrt{d}}{d} \rightarrow 0$, as $d \rightarrow \infty$.

The square of distances of data points **concentrate around an annulus** with radius $\approx d$!

To see it, simulate:

n : number of random points.



They really concentrate on their expectation $2d$!

To formalize it, we need concentration inequalities:

- ▶ Markov inequality: $x \geq 0$, $\mathbb{P}(x \geq a) \leq \frac{\mathbb{E}[x]}{a}$.
- ▶ Chebyshev's inequality: $\mathbb{P}(|x - \mathbb{E}[x]| \geq c) \leq \frac{\text{Var}(x)}{c^2}$.
- ▶ **the Law of Large Number.**

To formalize the concentration behavior of $\sum_{i=1}^d (y_i - z_i)^2$, we use the **Law of Large Number** (LLN): since $x_i := (y_i - z_i)^2$ are independent samples of a r.v x of bounded variance,

$$\mathbb{P}\left(\left|\frac{x_1 + x_2 + \cdots + x_n}{n} - \mathbb{E}[x]\right| \geq \epsilon\right) \leq \frac{\text{Var}(x)}{n\epsilon^2}.$$

Proof.

(Chebyshev's Inequality)

$$\begin{aligned}\mathbb{P}\left(\left|\frac{x_1 + x_2 + \cdots + x_n}{n} - \mathbb{E}[x]\right| \geq \epsilon\right) &\leq \frac{\text{Var}\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right)}{\epsilon^2} \\ &= \frac{\text{Var}(x_1 + x_2 + \cdots + x_n)}{n^2\epsilon^2} \\ &= \frac{\text{Var}(x)}{n\epsilon^2}.\end{aligned}$$



Application I: Vanishing Volume of the Unit Ball

- ▶ $\mathbf{x}$ with $x_i \sim N(0, \frac{1}{2\pi})$. Then the density in B_1 is lower-bounded by $\exp(-\pi)$.
- ▶ By **LLN**: $\|\mathbf{x}\|^2 = \sum_{i=1}^d x_i^2 = \Theta(d)$ with high probability.
- ▶ The probability that point $\mathbf{x}$ lies in the unit ball is **vanishingly small**.
- ▶ Since Probability Mass $= \int_{B_1} \text{Probability Density} \geq \exp(-\pi) \cdot \text{Volume}(B_1)$, the volume of the unit ball also vanishes as $d \rightarrow \infty$.

Remark: use probabilistic behaviours to learn geometric properties.

Application II

draw y, z with coordinates from $N(0, 1)$.

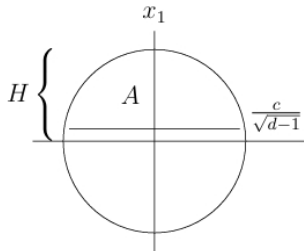
- ▶ $\|y\|^2 \approx d, \|z\|^2 \approx d$.
- ▶ $\|y - z\|^2 \approx 2d$.
- ▶ Thus $\|y - z\|^2 \approx \|y\|^2 + \|z\|^2$.
- ▶ **Pythagorean theorem** $\implies$ random d -dimensional y, z are approximately orthogonal.

Application II

draw y, z with coordinates from $N(0, 1)$.

- **Pythagorean theorem** $\implies$ random d -dimensional y, z are approximately orthogonal.

"Volume near the equator", we will formalize it.



Tail bounds for sums of r.v. with certain distributions.

Theorem 2.5 (Master Tail Bounds Theorem) *Let $x = x_1 + x_2 + \cdots + x_n$, where $x_1, x_2, \dots, x_n$ are mutually independent random variables with zero mean and variance at most σ^2 . Let $0 \leq a \leq \sqrt{2n}\sigma$. Assume that $|E(x_i^s)| \leq \sigma^2 s!$ for $s = 3, 4, \dots, \lfloor (a^2/4n\sigma^2) \rfloor$. Then,*

$$\text{Prob}(|x| \geq a) \leq 3e^{-a^2/(12n\sigma^2)}.$$

Tail bounds for sums of r.v. with certain distributions.

	Condition	Tail bound
Markov	$x \geq 0$	$\text{Prob}(x \geq a) \leq \frac{E(x)}{a}$
Chebyshev	Any x	$\text{Prob}(x - E(x) \geq a) \leq \frac{\text{Var}(x)}{a^2}$
Chernoff	$x = x_1 + x_2 + \dots + x_n$ $x_i \in [0, 1]$ i.i.d. Bernoulli;	$\text{Prob}(x - E(x) \geq \varepsilon E(x))$ $\leq 3e^{-c\varepsilon^2 E(x)}$
Higher Moments	r positive even integer	$\text{Prob}(x \geq a) \leq E(x^r)/a^r$
Gaussian Annulus	$x = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$ $x_i \sim N(0, 1); \beta \leq \sqrt{n}$ indep.	$\text{Prob}(x - \sqrt{n} \geq \beta) \leq 3e^{-c\beta^2}$
Power Law for x_i ; order $k \geq 4$	$x = x_1 + x_2 + \dots + x_n$ x_i i.i.d ; $\varepsilon \leq 1/k^2$	$\text{Prob}(x - E(x) \geq \varepsilon E(x))$ $\leq (4/\varepsilon^2 kn)^{(k-3)/2}$

Problems in the last week:

- Large deviations (tail bounds): If $a = t\sigma\sqrt{n}$ scales with $\sqrt{n}$, the Master Tail Bounds Theorem (Theorem 2.5 in the book) provides a non-asymptotic exponential bound:

$$P\left(\left|\frac{x}{\sigma\sqrt{n}}\right| \geq t\right) \leq 3e^{-\frac{t^2}{12}}.$$

This is a large deviation result, independent of n , and does not directly describe convergence to normality but ensures sub-Gaussian/exponential tail decay.

- For a fixed, how fast does $P(|x| \geq a)$ converge to 0, as $n \rightarrow \infty$? (The theorem does not tell us. Try CLT.)

The rate of convergence for the normalized sum $\frac{x}{\sigma\sqrt{n}}$ (where x is the sum of n i.i.d. random variables with mean 0 and variance σ^2) to the standard normal distribution:

Fixed deviation (CLT regime): If t is fixed and independent of n , the Central Limit Theorem (CLT) states that $\frac{x}{\sigma\sqrt{n}}$ converges in distribution to $\mathcal{N}(0,1)$. The Berry-Esseen theorem quantifies the rate of convergence in the Kolmogorov-Smirnov distance:

$$\sup_t \left| P\left(\frac{x}{\sigma\sqrt{n}} \leq t\right) - \Phi(t) \right| \leq \frac{C\rho}{\sigma^3\sqrt{n}},$$

where $\rho = \mathbb{E}[|X_i|^3]$, and C is a constant (typically ≤ 0.56). This gives a convergence rate of $\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$.

We are ready to derive the convergence rate of $P(|x| \geq a) \rightarrow 1$.

$$P(|x| \geq a) = P\left(\frac{x}{\sigma\sqrt{n}} \leq \frac{a}{\sigma\sqrt{n}}\right) \text{ Let } t = a/\sigma\sqrt{n}.$$

$$\sup_a \left| P\left(\frac{x}{\sigma\sqrt{n}} \leq \frac{a}{\sigma\sqrt{n}}\right) - \Phi\left(\frac{a}{\sigma\sqrt{n}}\right) \right| \leq \frac{C\rho}{\sigma^3\sqrt{n}},$$

$$\begin{aligned} \Phi\left(\frac{a}{\sigma\sqrt{n}}\right) - \Phi\left(-\frac{a}{\sigma\sqrt{n}}\right) - O\left(\frac{1}{\sqrt{n}}\right) &\leq P\left(-\frac{a}{\sigma\sqrt{n}} \leq \frac{x}{\sigma\sqrt{n}} \leq \frac{a}{\sigma\sqrt{n}}\right) \\ &\leq \Phi\left(\frac{a}{\sigma\sqrt{n}}\right) - \Phi\left(-\frac{a}{\sigma\sqrt{n}}\right) + O\left(\frac{1}{\sqrt{n}}\right). \end{aligned} \tag{3}$$

$$\Phi(x) = \frac{1}{2}[1 + \text{Erf}(x/\sqrt{2})]$$

$$\text{Erf}(x) = \int_0^x e^{-t^2} dt \approx \frac{2}{\sqrt{\pi}} \text{ as } x \rightarrow 0.$$

$$\text{For } n \text{ large, } \Phi\left(\frac{a}{\sigma\sqrt{n}}\right) - \Phi\left(-\frac{a}{\sigma\sqrt{n}}\right) = \text{Erf}\left(\frac{a}{\sigma\sqrt{2n}}\right) \sim \frac{2}{\sqrt{\pi}} \frac{a}{\sigma\sqrt{2n}}$$

$$\text{In conclusion, } P(|x| \geq a) = 1 - O\left(\frac{1}{\sqrt{n}}\right)$$

Outline

Properties of High-Dimensional space

Properties of the Unit Ball

Generating Points Uniformly at Random from a Ball

Random Projection and JL Lemma

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Geometry of High Dimensions

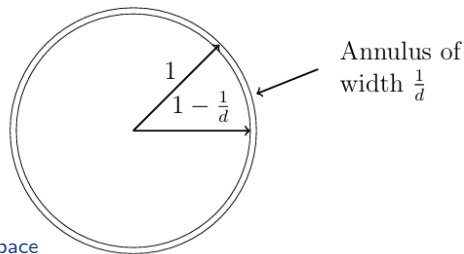
- Most of the volume of the high-dimensional objects is near its surface:

$$\frac{V((1 - \epsilon)A)}{V(A)} = (1 - \epsilon)^d \leq e^{-\epsilon d}. \quad (4)$$

Fixing ϵ and letting $d \rightarrow \infty$, the above quantity rapidly approaches zero.

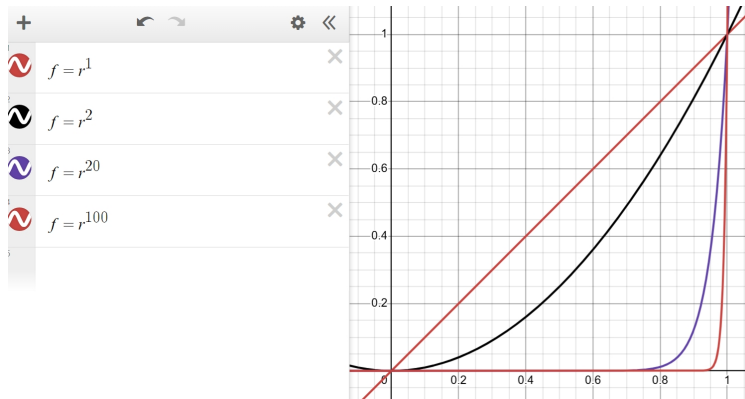
In particular, consider $\epsilon = O(\frac{1}{d})$.

Let B_1 denote the unit ball. At least $1 - e^{-\epsilon d}$ fraction of the volume is in $B_1 / (1 - \epsilon)B_1$



- Let $0 < r < 1$, the fraction of volume in B_r over the total volume:

$$\frac{\text{Vol}(rB_1)}{\text{Vol}(B_1)} = \frac{r^d \text{Vol}(B_1)}{\text{Vol}(B_1)} = r^d.$$



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$$\begin{aligned} V(d) &= \int_{S^{d-1}} \int_{r=0}^1 r^{d-1} dr d\Omega \\ &= \int_{S^{d-1}} d\Omega \int_{r=0}^1 r^{d-1} dr \\ &= \frac{A(d)}{d}. \end{aligned} \tag{5}$$

To evaluate $A(d)$, compute $I(d) = \int_{\mathbb{R}^d} e^{-\|x\|^2} dx$ in two ways:

- ▶ Cartesian coordinates: $I(d) = (\int_{-\infty}^{\infty} e^{-x^2})^d = \pi^{d/2}$.
- ▶ Polar coordinates: $I(d) = \int_{S^{d-1}} d\Omega \int_{r=0}^1 e^{-r^2} r^{d-1} dr = A(d) (\frac{1}{2} \Gamma(\frac{d}{2}))$, since

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt, \Gamma(n) = (n-1)!$$

$$\implies A(d) = \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})} \text{ and } V(d) = \frac{2\pi^{\frac{d}{2}}}{d\Gamma(\frac{d}{2})}.$$

$$A(d) = \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})} \text{ and } V(d) = \frac{2\pi^{\frac{d}{2}}}{d\Gamma(\frac{d}{2})}.$$

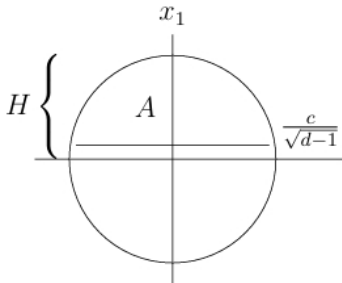
► factorial $>$ exponential $\implies \lim_{d \rightarrow \infty} V(d) = 0.$

Volume Near the Equator

For any unit-length vector v defining **north**, most of the volume of the unit ball lies in the thin slab of points whose dot product with v has magnitude $O(\frac{1}{\sqrt{d}})$.

Theorem 1.

For $c \geq 1$ and $d \geq 3$, at least a $1 - \frac{2}{c}e^{-\frac{c^2}{2}}$ fraction of the volume of the d -dimensional unit ball has $|x_1| \leq \frac{c}{\sqrt{d-1}}$.



(For a region, the Porportion of the Volumn = the Probability Mass for Uniform Distribution)

Considering drawing n points $x_1, x_2, \dots, x_n$ at random from the unit ball. With probability $1 - O(\frac{1}{n})$,

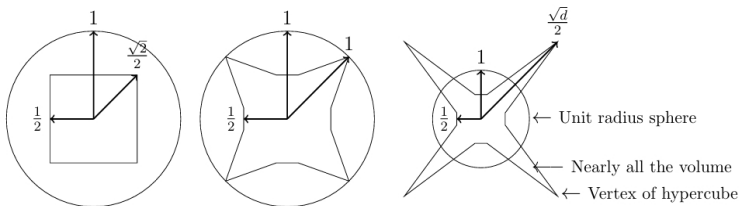
1. $\|x_i\| \geq 1 - \frac{2\ln n}{d}$ for all i , and
2. $|x_i \cdot x_j| \leq \frac{\sqrt{6\ln n}}{\sqrt{d-1}}$ for all $i \neq j$.

Proof. 1. follows from equation 4: $\frac{V((1-\epsilon)A)}{V(A)} \leq e^{-\epsilon d}$.

2. follows from Theorem 1 for individual arguments and taking a union bound.

How it can be that nearly all the points in the unit ball are very close to the surface and yet at the same time nearly all points are in a box of side-length $\Theta(\frac{\ln d}{d-1})$ $O(\sqrt{\frac{\ln d}{d-1}})$.

- Points on the surface satisfy $x_1^2 + x_2^2 + \dots + x_d^2 = 1$, so for each coordinates i a typical value will be $\pm O(\frac{1}{\sqrt{d}})$.
- Illustration:



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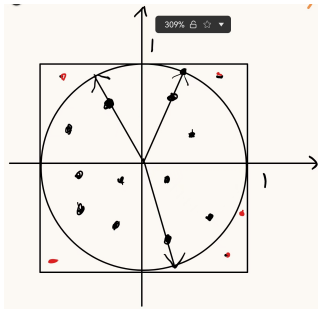
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On the Surface

When $d = 2$,



► How about d is large?

- **Surface**: Spherical Gaussian distribution + Normalizing
- **Surface+interior**: Scale the point on the surface at random.

On the Surface

When d is large,

- ▶ $\mathbf{x} = (x_1, \dots, x_d)^T \sim N(0, I_d)$, i.e. $(\frac{1}{\sqrt{2\pi}})^d \exp(-\frac{\|\mathbf{x}\|^2}{2})$.
- ▶ Normalize to a unit vector $\frac{\mathbf{x}}{\|\mathbf{x}\|}$.

Over the Unit Ball

When d is large, generate a point y over B_1 (surface and interior):

- ▶ Scale the point $\frac{x}{\|x\|}$ generated uniformly on the surface by a random variable ρ .
- ▶ Compute the **distribution function** of $\rho \in [0, 1]$:

$$F_\rho(r) = \mathbb{P}(\rho \leq r) = \frac{r^d V(d)}{V(d)} = r^d. \quad (6)$$

Therefore pick ρ with density $f_\rho(r) = \frac{d}{dr} F_\rho(r) = dr^{d-1}$.

- ▶ Set $y = \rho \frac{x}{\|x\|}$.

The d -dimensional spherical Gaussian with zero mean and variance σ^2 in each coordinate has density function

$$p(\mathbf{x}) = \left(\frac{1}{\sqrt{2\pi}}\right)^d \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma^2}\right) \quad (7)$$

- ▶ the probability mass in the unit ball is almost zero, since $V(d)$ is negligible.
- ▶ By previous analysis, the radius of the ball need to be nealy $\sqrt{d}$ before there is a significant volumn and hence significant probability mass.

Gaussian Annulus Theorem describe this fact.

Gaussian Annulus Theorem

For a d -dimensionality spherical Gaussian with unit variance in each direction, for any $\beta \leq \sqrt{d}$, all but at most $3e^{-c\beta^2}$ of the probability mass lies within the annulus $\sqrt{d} - \beta \leq \|x\| \leq \sqrt{d} + \beta$, where c is a fixed positive constant.

Proof. see the textbook.

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Nearest Neighbor Search:

Given a database of n points in $\mathbb{R}^d$, find the nearest or approximately nearest database point to the query point.

When d is large, it could cost more than expected.

- ▶ **dimension reduction:** projects the database points to a k -dimensional space with $k \ll d$ can be very useful so long as the relative distances between points are approximately preserved.

Johnson-Lindenstrass Lemma tells us that to put n vectors, we only need $O(\ln n)$ dimensions, while keep the relative distances between points are approximately preserved.

Projection Function

Pick k vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in \mathbb{R}^d$ from $N(0, I_d)$, for any vector $\mathbf{v}$, the projection $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is:

$$f(\mathbf{v}) := (\mathbf{u}_1 \cdot \mathbf{v}, \mathbf{u}_2 \cdot \mathbf{v}, \dots, \mathbf{u}_k \cdot \mathbf{v}) \quad (8)$$

$$\|f(\mathbf{v})\| \approx \sqrt{k} \|\mathbf{v}\|.$$

To compute $\|\mathbf{v}_1 - \mathbf{v}_2\|$, it suffices to compute $\|f(\mathbf{v}_1 - \mathbf{v}_2)\| = \|f(\mathbf{v}_1) - f(\mathbf{v}_2)\|$.

Theorem 2.10 (The Random Projection Theorem) *Let $\mathbf{v}$ be a fixed vector in $\mathbf{R}^d$ and let f be defined as above. There exists constant $c > 0$ such that for $\varepsilon \in (0, 1)$,*

$$\text{Prob} \left(\left| |f(\mathbf{v})| - \sqrt{k}|\mathbf{v}| \right| \geq \varepsilon \sqrt{k}|\mathbf{v}| \right) \leq 3e^{-ck\varepsilon^2},$$

where the probability is taken over the random draws of vectors $\mathbf{u}_i$ used to construct f .

Proof. Use the Gaussian Annulus Theorem.

Theorem 2.11 (Johnson-Lindenstrauss Lemma) For any $0 < \varepsilon < 1$ and any integer n , let $k \geq \frac{3}{c\varepsilon^2} \ln n$ with c as in Theorem 2.9. For any set of n points in R^d , the random projection $f : R^d \rightarrow R^k$ defined above has the property that for all pairs of points $\mathbf{v}_i$ and $\mathbf{v}_j$, with probability at least $1 - 3/2n$,

$$(1 - \varepsilon)\sqrt{k} \|\mathbf{v}_i - \mathbf{v}_j\| \leq \|f(\mathbf{v}_i) - f(\mathbf{v}_j)\| \leq (1 + \varepsilon)\sqrt{k} \|\mathbf{v}_i - \mathbf{v}_j\|.$$

Proof. Using The Random Projection Theorem, take $k \geq \frac{3 \ln n}{c\varepsilon^2}$ to control the union bound $O(n^2)(3e^{-cke^2})$

Johnson-Lindenstrauss Lemma tells us that put n vectors, we only need $O(\ln n)$ dimensions.

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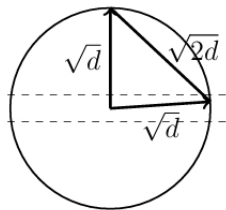
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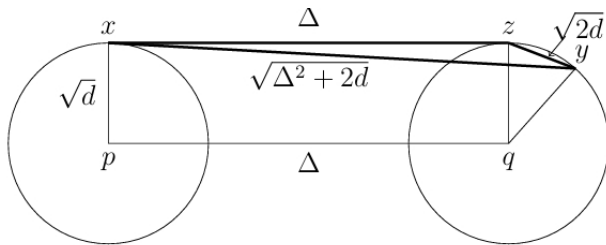
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(a)



(b)

► x, y are from one spherical same unit variance Gaussian, $\|x - y\|^2 = 2d \pm O(\sqrt{d})$.

► x, y are from two Gaussians,

$$\|x - y\|^2 \approx \Delta^2 + \|z - q\|^2 + \|q - y\|^2 = \Delta^2 + 2d \pm O(\sqrt{d}).$$

Let $2d \pm O(\sqrt{d}) \leq \Delta^2 + 2d \pm O(\sqrt{d})$,

$$\implies \Delta \in \omega(d^{1/4}).$$

Question: what if we have n points and want to correctly separate all of them with high probability?

In Gaussian Annulus Theorem, take $\beta = O(\sqrt{\log n})$ (instead of $O(1)$).

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Given n samples x_i from $g := N(\mu, \sigma^2)$ (parameters unknown), the probability density for picking these points when sampling according to g is given by

$$L(\mu, \sigma) := f(x_1, \dots, x_n; \mu, \sigma) = \frac{1}{(2\pi\sigma^2)^{nd/2}} \exp\left(-\frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + \dots + (x_n - \mu)^2}{2\sigma^2}\right) \quad (9)$$

Maximum Likelihood Estimator (MLE) of g given the samples x_i is the g that maximizes the above probability density $L(\mu, \sigma)$. For $\hat{\mu}_{MLE}$,

$$\begin{aligned}\hat{\mu}_{MLE} &= \arg \max_{\mu} \exp\left(-\frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + \cdots + (x_n - \mu)^2}{2\sigma^2}\right) \\ &= \arg \min_{\mu} (x_1 - \mu)^2 + (x_2 - \mu)^2 + \cdots + (x_n - \mu)^2 \\ &= \frac{1}{n}(x_1 + x_2 + \cdots + x_n).\end{aligned}\tag{10}$$

To determine $\hat{\sigma}_{MLE}^2$, set μ to be the true centroid. Denote $a = \sum_{i=1}^n (x_i - \mu)^2$

Take a log: $l(\sigma) = \ln L(\sigma) = -a \frac{1}{2\sigma^2} - \frac{nd}{2} \ln(2\pi\sigma^2)$.

Set $l'(\sigma) = a \frac{1}{\sigma^3} - \frac{nd}{2} \frac{2\pi \cdot 2\sigma}{2\pi\sigma^2} = 0$.

$$\hat{\sigma}_{MLE}^2 = \frac{a}{nd}.$$